DEPARTMENT
OF
ELECTRICAL
ENGINEERING

Computer

Programming

- Assignment

#2

Assignment Solutions

Question 1: Max, Min, and Average of 10 integers.

```
#include <iostream>
using namespace std;

int main() {
    int arr[10], sum = 0;
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
        sum += arr[i]; // Summing for average calculation
    }
    int maxVal = arr[0], minVal = arr[0];
    for(int i = 1; i < 10; i++) {
        if(arr[i] > maxVal) maxVal = arr[i]; // Compare and update max
        if(arr[i] < minVal) minVal = arr[i]; // Compare and update min
    }
    return 0;
}
```

Question 2: Reverse an array of 10 integers.

```
#include <iostream>
#include <algorithm>
using namespace std;

int main() {
    int arr[10];
    for(int i = 0; i < 10; i++) cin >> arr[i];
    // Reverse the array using in-place swapping
    for(int i = 0; i < 5; i++) swap(arr[i], arr[9-i]);
    return 0;
}
```

Question 3: 3x3 Matrix Addition.

```
#include <iostream>
using namespace std;

int main() {
    int A[3][3], B[3][3], Sum[3][3];
    for(int i=0; i<3; i++) {
        for(int j=0; j<3; j++) {
            Sum[i][j] = A[i][j] + B[i][j]; // Element-wise addition of matrices
        }
    }
    return 0;
}
```

Question 4: Sort 5 names in alphabetical order.

```
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;

int main() {
    string names[5];
    for(int i = 0; i < 5; i++) cin >> names[i];
    sort(names, names + 5); // Lexicographical sort
    return 0;
}
```

Question 5: Power load tracker (24 hours).

```
#include <iostream>
using namespace std;

int main() {
    double load[24], peak = 0;
    for(int i = 0; i < 24; i++) {
        cin >> load[i];
        if(load[i] > peak) peak = load[i]; // Identify peak consumption
    }
    return 0;
}
```

Question 6: Student Management System.

```
#include <iostream>
#include <string>
using namespace std;

struct Student { string name; int marks; };
int main() {
    Student s[5];
    int topIdx = 0;
    for(int i=0; i<5; i++) {
        cin >> s[i].name >> s[i].marks;
        if(s[i].marks > s[topIdx].marks) topIdx = i; // Compare marks for topper
    }
    return 0;
}
```

Question 7: Solar energy generation analysis.

```
#include <iostream>
using namespace std;

int main() {
    double solar[24], maxG = 0;
    int hour = 0;
    for(int i=0; i<24; i++) {
        cin >> solar[i];
        if(solar[i] > maxG) { maxG = solar[i]; hour = i; } // Log hour of max generation
    }
    return 0;
}
```

Question 8: Battery voltage monitoring.

```
#include <iostream>
using namespace std;

int main() {
    double v;
    for(int i=0; i<20; i++) {
        cin >> v;
        if(v < 3.0 || v > 4.2) cout << "Warning!"; // Critical voltage threshold check
    }
    return 0;
}
```

Question 9: Smart grid demand analyzer.

```
#include <iostream>
using namespace std;

int main() {
    double grid[4][7], dayT[7] = {0};
    for(int i=0; i<4; i++) {
        for(int j=0; j<7; j++) {
            cin >> grid[i][j];
            dayT[j] += grid[i][j]; // Column-wise sum for daily totals
        }
    }
    return 0;
}
```

Question 10: Attendance percentage calculation.

```
#include <iostream>
using namespace std;

int main() {
    int att, count = 0;
    for(int i=0; i<35; i++) {
        cin >> att;
        if(att == 1) count++; // Increment present count for '1'
    }
    return 0;
}
```